

ICT1242 Web Development Practicum

Mini Project Proposal

Project Title

Hospital Appointment Admission and Billing System

Scenario 4

Group 05

Submitted by TG2095 - Hasaranga K.H.M
 TG2096 - Himansana N.V.S
 TG2097 - Nirmal I.D.N.S
 TG2112 - Sarathchandra G.M.S
 TG2086 - Minula Ishan
 TG2088 - Dasula Wosada
 TG2127 - Sadaru D
 TG2080 - Appuhami I.A.S.S
 TG2136 - Hirunika P.S.R

1. Project Title

Scenario Number: 4

Hospital Appointment, Admission, and Billing System

2. Group Details

Group Number: 05

#	Student ID	Student Name	Role
1	TG2095	Hasaranga K.H.M	Team Leader / Backend Developer (Login, Sessions, Dashboard)
2	TG2096	Himansana N.V.S	Backend Developer – Appointment Booking System
3	TG2097	Nirmal I.D.N.S	Frontend Developer – Doctor Schedule & Shared UI (CSS)
4	TG2112	Sarathchandra G.M.S	Frontend Developer – Ward Admission & Patient Records UI
5	TG2086	Minula Ishan	JavaScript Developer – Form Validation & DOM Manipulation
6	TG2088	Dasula Wosada	JavaScript Developer – Search, Filter & Dynamic Tables
7	TG2127	Sadaru D	Database Designer – MySQL Schema & All SQL Queries
8	TG2080	Appuhami I.A.S.S	Backend Developer – Treatment Records & Billing Module
9	TG2136	Hirunika P.S.R	Backend Developer – Patient Registration & System Testing / Documentation

3. Introduction

Private hospitals today are facing serious challenges in their operations, particularly in managing the flow of patients through various sections of the hospitals. Presently, hospitals are using manual records and paper-based appointment books, which are leading to serious problems in managing patient appointments. This situation reduces the quality of service provided to patients.

The Hospital Appointment, Admission, and Billing System is an information system developed as part of the ICT 1242 Web Development Practicum. It is a web-based information system designed to solve the problems faced by private hospitals in managing their operations through a centralized system.

The system is designed to meet the needs of various user groups in a private hospital. It supports four user roles, namely:

- Patients – who can register, book appointments, book emergency ambulances, and view their appointments and bills.
- Reception Staff – who can manage walk-in appointments for the outpatient department, admit patients to rooms, and manage billing, payment, and discharge.
- Doctors – who can view their appointments, make diagnoses, prescribe drugs, and book operations.
- Administrators – who can manage all system activities, including managing staff accounts, viewing revenue, and managing ward occupancy.

The system is developed with seven integrated modules, namely Appointment Management, Emergency Ambulance, Ward Room Management, Theatre Operation Management, Pharmacy Drug Management, Billing and Payment, and Admin Dashboard. These modules use a single relational database management system (MySQL).

The system is developed using HTML, CSS, Vanilla JavaScript, PHP, and MySQL, which are mandated by the course requirements of ICT 1242.

4. Functional Requirements

The system provides the following functional capabilities across all modules:

4.1 User Registration and Authentication

- User registration with email, password, and role selection (Patient, Doctor, Reception, Pharmacist, Admin).
- Secure login using PHP sessions and decrypt-hashed passwords.
- Role-based dashboard routing — each role sees only their relevant pages and functions.
- Session management and logout with session destruction.
- Admin can activate or deactivate any user account.

4.2 Patient Management

- Register new patients with full personal and medical details (NIC, DOB, gender, blood type, address, emergency contact).
- Search and filter patient records by name, NIC, or registration date.
- Update existing patient profile information.
- View complete patient history including past appointments, admissions, treatments, and invoices.

4.3 Appointment Management (Online + OPD Sync)

- Patients can book appointments online by selecting a doctor, date, and available time slot.
- Reception staff can create walk-in OPD appointments using the same shared calendar.
- System prevents double-booking — both online and OPD channels share one live appointment table.
- Unique appointment reference number auto-generated on confirmation.
- Patients can reschedule or cancel appointments from their dashboard.
- Filter appointments by doctor, date, or status (Pending / Confirmed / Completed / Cancelled).
- Admin and reception can edit or delete any appointment record.

4.4 Emergency Ambulance Dispatch

- Patients can submit an emergency request from the hospital emergency contact number.
- A unique emergency ticket number is generated.
- Dispatcher views all incoming requests in real time on the admin panel.
- Dispatcher assigns an available ambulance to the request; status updates through: Pending → Dispatched → En Route → Arrived.
- The emergency ward is automatically notified in the system upon dispatch.

4.5 Ward and Room Management

- Four room categories available: Semi-Private, Private, Children's Room, and ICU, each with real-time bed availability tracking.
- Reception admits patients to an appropriate room, recording admission date, assigned doctor, and dietary requirements.
- System manages two meal plan types: hospital-standard diet and doctor-prescribed special diet.
- Reception can discharge patients, automatically updating room availability.

4.6 Theatre (Operation) Management

- Admin schedules operations by selecting the patient, lead surgeon, anaesthesiologist, operation type, theatre number, and date/time.
- Theatre calendar prevents double-booking of the same theatre at the same time.
- Post-operation, the patient is automatically transferred to the appropriate recovery ward.
- Theatre charges (surgical fee, theatre use fee, anesthesia fee) are automatically added to the patient's billing invoice.
- Maternity operations are linked to newborn record creation.

4.7 Pharmacy and Drug Management

- Pharmacist maintains a drug catalogue — add, edit, and deactivate drugs with name, category, stock quantity, unit price, and reorder level.
- System generates low-stock alerts when quantity falls below the reorder level.
- Doctors issue prescriptions linked to a patient's treatment record; prescriptions appear in the pharmacist's dispensing queue.
- On dispensing, stock quantity is automatically deducted and a dispensing record is created for audit.
- Dispensed medications are automatically added as line items to the patient's billing invoice.

4.8 Billing and Payment

- Patient invoices are auto-generated, aggregating charges from all modules: consultation fees, room/ward charges (per day), theatre fees, pharmacy charges, and any miscellaneous items.
- Supports two payment modes: Full Payment (complete outstanding balance) and Advance Payment (partial payment during admission with balance tracked).
- Each payment is recorded with date, amount and payment method.
- Invoice balance updates in real time after each payment.
- Payment receipt generated for every payment; final settlement receipt on discharge.
- Reception can add, edit, and delete billing line items manually if needed.

4.9 Admin Dashboard and Reporting

- Live system overview: today's total appointments, currently admitted patients, pending bills, and active emergency requests.
- Revenue summary: daily, weekly, and monthly revenue from billing records.
- Ward occupancy: room-by-room availability across all room types.
- Appointment report: filter by doctor, date range, or status.
- Staff management: add, edit, and deactivate reception and pharmacy staff accounts.

5. Proposed System Modules / Pages

The system is organized into the following pages and modules. All pages share a common navigation header and role-based sidebar.

Page / Module	Accessible By	Key Features
Login & Register	All Users	Role-based login, session start, registration form, logout
Admin Dashboard	Admin	Live stats (appointments, admissions, revenue, emergencies), ward occupancy
Patient Management	Reception, Admin	Register, search, view, edit, delete patient records; full patient history view
Appointment Booking	Patient, Reception	Online & OPD booking, doctor/date/slot selection, reschedule, cancel, status filter
Doctor Schedule	Doctor, Admin	Set available slots per day, block slots, view weekly calendar
Emergency Dispatch	Patient (request), Admin/Dispatcher (manage)	GPS-based ambulance request, dispatcher view, assign ambulance, status tracking
Ward & Room Management	Reception, Admin	Room availability, patient admission, dietary notes, discharge, maternity/newborn link
Theatre Scheduling	Admin, Doctor	Schedule operations, theatre calendar, prevent double-booking, post-op ward transfer
Treatment Records	Doctor, Admin	Diagnosis, clinical notes, follow-up date, linked prescriptions per appointment
Pharmacy Management	Pharmacist, Doctor, Admin	Drug catalogue, low-stock alerts, prescription queue, dispensing, auto-billing
Billing & Payments	Reception, Admin	Auto-generated invoices, full/advance payments, receipts, balance tracking
Patient Portal	Patient	View own appointments, treatment history, invoices, emergency request status
Reports	Admin	Revenue summary, appointment stats, payment tracking, staff activity audit log

6. Student Contribution

Each group member has been assigned an equal and meaningful area of responsibility. All members contribute to both frontend and backend aspects of their assigned module.

Student ID	Responsibility	Detailed Tasks
TG2095	Team Lead + Backend Developer	Create and manage the GitHub repository and branch structure Build login.php, register.php, session management (session_check.php) Develop Admin Dashboard page with live stats Write db_config.php and shared PHP includes (header, footer, sidebar) Coordinate team daily check-ins and integration of all modules
TG2096	Backend Developer – Appointments	Build appointment booking system: online + OPD walk-in unified flow PHP: book_appointment.php, cancel, reschedule, appointments.php list SQL: INSERT, SELECT, UPDATE, DELETE for appointments table Prevent double-booking logic in PHP + MySQL
TG2097	Frontend Developer – UI & CSS	Design and build the complete CSS system: main.css, forms.css, tables.css, dashboard.css Build shared header.php, sidebar.php (role-based nav), footer.php Build Doctor Schedule page HTML + PHP frontend Ensure consistent, responsive layout across all pages
TG2112	Frontend Developer – Admission UI	Build Ward Admission page HTML/CSS: admission form, room selection, dietary notes Build Patient Records HTML pages: detailed view, history timeline Build Maternity / Newborn registration form UI CSS for tables.css and ward-specific components
TG2086	JavaScript Developer – Validation & DOM	Write validate.js: form validation for all pages (required fields, formats, date checks) Write dom.js: show/hide form sections, success/error notification banners Implement JS arrays and objects to manage form data before submission Handle all keyboard events (Enter key, search-on-type triggers)

TG2088	JavaScript Developer – Search & Tables	Write search.js: live search across patient, appointment, drug tables as user types Write tables.js: dynamically add/remove billing line item rows via DOM Write filter.js: filter records by status, date range, doctor using JS arrays of objects Implement renderTable() and clearForm() reusable JS functions
TG2127	Database Designer – MySQL	Design and write the complete MySQL schema: hospital_db.sql (17 tables) Define all primary keys, foreign keys, indexes, and ENUM constraints Write queries.sql: all INSERT, SELECT, UPDATE, DELETE queries used by the system Document all table relationships and ensure referential integrity
TG2080	Backend Developer – Billing & Treatment	Build treatment.php: record diagnosis, notes, followup date per appointment Build billing.php: auto-generate invoice from all module charges Build billing_report.php: advance payment recording, balance tracking, receipts SQL: billing_invoices, billing_items, payments table operations
TG2136	Backend Developer – Patients & Testing	Build patients.php: patient CRUD — register, search, edit, delete Build register_patient.php: full patient registration form with validation Write and submit the project proposal document Conduct full system testing across all four user roles; document test results

7. Expected Database Entities

The system database is named `hospital_db` and contains 17 tables, organized across 7 functional groups. This exceeds the minimum requirement of 4 tables specified in the project brief. All tables are implemented in MySQL and accessed through PHP PDO.

#	Table Name	Module Group	What It Stores
1	users	User & Core	Login credentials and role for all system users (Patient, Doctor, Reception, Pharmacist, Admin)
2	patients	User & Core	Extended patient profile: NIC, DOB, gender, blood type, emergency contact
3	doctors	User & Core	Doctor profile: specialization, qualifications, license number, consultation fee
4	doctor_schedules	Appointment	Recurring available time slots per doctor per day of week
5	appointments	Appointment	All appointments — online and OPD walk-in — with status and source tracking
6	treatment_records	Appointment	Doctor diagnosis, clinical notes, and follow-up date per appointment
7	emergency_requests	Emergency	Patient ambulance requests with GPS coordinates, phone, status, and assigned ambulance
8	ambulances	Emergency	Hospital ambulance fleet: vehicle number, driver details, availability status
9	rooms	Ward & Room	All hospital rooms by type (Semi-Private, Private, Children, ICU) with bed availability and daily rate
10	admissions	Ward & Room	Patient admissions: room, doctor, dates, dietary requirements, discharge status
11	newborn_records	Ward & Room	Newborn data linked to mother and labour operation: DOB, weight, health status, assigned room
12	theatre_operations	Theatre	Scheduled operations: surgeon, anaesthetist, operation type, theatre number, pre/post-op notes
13	pharmacy_drugs	Pharmacy	Drug catalogue: name, category, unit, price, stock quantity, reorder level
14	prescriptions	Pharmacy	Doctor-issued prescriptions: drug, dosage, frequency, dispensing status and pharmacist
15	billing_invoices	Billing	Master invoice per patient: total amount, paid amount, balance, and settlement status
16	billing_items	Billing	Individual invoice line items: consultation, room, theatre, pharmacy, lab, miscellaneous charges
17	payments	Billing	Payment transactions: amount, full/advance type, method, receipt number, receiving staff

Key Table Relationships

- users is the central identity table — one user record maps 1:1 to either a patient or doctor profile.
- patients → appointments → treatment_records → prescriptions forms the core outpatient data chain.
- patients → admissions → billing_invoices forms the core inpatient billing chain.
- theatre_operations links to patients (surgery), doctors (surgeon + anaesthetist), rooms (post-op ward), and newborn_records (maternity).
- billing_invoices → billing_items (auto-populated by appointment, admission, theatre, and pharmacy modules) → payments supports the complete billing lifecycle.
- emergency_requests links to patients (if registered), ambulances (on dispatch), and users (dispatcher).

End of Proposal — Group 05 — ICT1242 Web Development Practicum